GENERAL INFORMATION

DIAGNOSTIC TROUBLE CODE INDEX - DTC: IMAGE PROCESSING MODULE (IPMA) [G2046450]

DESCRIPTION AND OPERATION

IMAGE PROCESSING MODULE (IPMA)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- Guided Diagnostics must be followed and the repair advised by the process completed. Failure to do so may result in the rejection of any warranty claim made.
- If a control module or a component is not to specification and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval program is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter that has the resolution ability to view 3 decimal places. For example, on the 2 volts range can measure 1mV or 2 K Ohm range can measure 1 Ohm. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If Diagnostic Trouble Code(s) are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always inspect for loose connections and corroded terminals.
- Check the JLR claims submission system for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and complete the recommendations as required.

The table below lists all Diagnostic Trouble Code(s) that could be set in the Image Processing Module (IPMA). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual. For additional information, refer to: Headlamps (417-01 Exterior Lighting, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1206-53	Crash Occurred - Deactivated	▪ Crash signal received	▪ Using the Jaguar Land Rover approved diagnostic equipment, check the restraints control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B151F-18	Auxiliary Cooling Fan - Circuit current below threshold	▪ Image processing module cooling fan circuit open circuit, high resistance ▪ Image processing module internal failure	**NOTE:** Only replace the image processing module (IPMA) if DTC C1001-49 is also set ▪ Refer to the electrical circuit diagrams and check the image processing module cooling fan circuit for open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest. If the fault persists, install a new image processing module
B151F-19	Auxiliary Cooling Fan - Circuit current above threshold	▪ Auxiliary cooling fan circuit short circuit to ground, short circuit to power ▪ Battery/charging system fault	**NOTE:** Only replace the image processing module (IPMA) if DTC C1001-49 is also set ▪ Refer to the electrical circuit diagrams and check the rear auxiliary cooling fan circuit for short circuit to ground, short circuit to power. Repair the wiring harness as necessary ▪ Refer to the relevant section of the workshop manual and test the battery and charging system
B151F-31	Auxiliary Cooling Fan - No signal	▪ Image processing module cooling fan circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Image processing module internal failure	**NOTES:** ▪ Only replace the image processing module (IPMA) if DTC C1001-49 is also set ▪ This DTC is set if no feedback from the fan signal is received for more than 5 seconds ▪ Refer to the electrical circuit diagrams and check the image processing module cooling fan circuit for open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest. If the fault persists, install a new image processing module
B151F-91	Auxiliary Cooling Fan - Parametric	▪ Image processing module (IPMA) is not configured correctly	▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest. If the fault persists, reconfigure the image processing module with the latest level software. Clear the Diagnostic Trouble Code(s), road test the vehicle and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1DA9-92	Autonomous Emergency Braking Function - Performance or incorrect operation	▪ Image processing module is not configured correctly ▪ Image processing module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Image processing module internal failure	**NOTE:** In order to calibrate the image processing module, the road test must run for at least 5 minutes at speeds above 60km/h. If the vehicle speed drops below 30km/h, calibration will resume when the speed reaches 60km/h ▪ Using the Jaguar Land Rover approved diagnostic equipment, reconfigure the Image processing module with the latest level software. Clear the Diagnostic Trouble Code(s), road test the vehicle and retest ▪ If the DTC returns, refer to the electrical circuit diagrams and check the image processing module circuits for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ If the fault persists, install a new image processing module
C1001-49	Vision System Camera - Internal electronic failure	▪ Image processing module is not configured correctly ▪ Image processing module internal failure	▪ Using Jaguar Land Rover approved diagnostic equipment, reconfigure the image processing module with the latest level software ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest. If the fault persists, install a new image processing module
C1001-57	Vision System Camera - Invalid/incomplete software component	▪ Image processing module is not configured correctly ▪ Image processing module is not configured correctly	▪ Using Jaguar Land Rover approved diagnostic equipment, reconfigure the image processing module with the latest level software ▪ Using the Jaguar Land Rover approved diagnostic equipment, reconfigure the image processing module with the latest level software
C1001-78	Vision System Camera - Alignment or adjustment incorrect	▪ Image processing module camera misaligned	**NOTES:** ▪ To calibrate the image processing module, perform a road test for at least 5 minutes at speeds greater than 60 km/h. If the vehicle speed falls below 30 km/h, calibration will resume when the speed reaches 60 km/h ▪ If Diagnostic Trouble Code(s) U2300-51 or U2300-55 are set, check the symptoms for these Diagnostic Trouble Code(s) first, rectify the fault, clear Diagnostic Trouble Code(s) and retest before running routine - Align Camera Aim (0x800C) ▪ Make sure the image processing module is securely mounted in the correct position. Rectify as required ▪ Using the Jaguar Land Rover approved diagnostic equipment, run the routine - Align Camera Aim (0x800C). Clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1001-98	Vision System Camera - Component or system over temperature	• Image processing control module internal temperature above threshold	• Allow the vehicle to cool. Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest. If the fault persists, install a new image processing control module
U0001-81	High Speed CAN Communication Bus - Invalid serial data received	• Invalid data received from another control module through the high speed CAN bus (chassis)	• Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
U0001-82	High Speed CAN Communication Bus - Alive/sequence counter incorrect / not updated	• Invalid data received from another control module through the high speed CAN bus (chassis)	**NOTE:** The generation of DTC-U0001-82 (Alive Counter not updated) by the IPMA ECU is due to expected boot delays within the InControl© Touch software and must not be taken as grounds for the exchange of either IPMA or InControl© Touch hardware • Please ignore this DTC
U0001-83	High Speed CAN Communication Bus - Value of signal protection calculation Incorrect	• Invalid data received from another control module through the high speed CAN bus (chassis)	• Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
U0001-87	High Speed CAN Communication Bus - Missing message	• Missing message from another control module through the high speed CAN bus (chassis)	• Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the missing message source control module. Check the relevant control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
U0001-88	High Speed CAN Communication Bus - Bus off	• High speed CAN bus (chassis) circuit short circuit to ground, short circuit to power, open circuit, high resistance	• Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
U0002-82	High Speed CAN Communication Bus Performance - Alive/sequence counter incorrect / not updated	• Invalid data received from another control module through the high speed CAN bus (chassis)	• Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
U0300-00	Internal Control Module Software Incompatibility - No sub type information	• Car configuration file mismatch with vehicle specification	• Using the Jaguar Land Rover approved diagnostic equipment, check and update the car configuration file as necessary. Clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U2300-51	Central Configuration - Not programed	▪ Car configuration file mismatch with vehicle specification	**NOTE:** After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the Diagnostic Trouble Code(s) ▪ Using the Jaguar Land Rover approved diagnostic equipment, check and update the car configuration file as necessary. Clear the Diagnostic Trouble Code(s) and retest
U2300-54	Central Configuration - Missing calibration	▪ Car configuration file mismatch with vehicle specification	**NOTE:** After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the Diagnostic Trouble Code(s) ▪ Using the Jaguar Land Rover approved diagnostic equipment, check and update the car configuration file as necessary. Clear the Diagnostic Trouble Code(s) and retest
U2300-55	Central Configuration - Not configured	▪ Car configuration file mismatch with vehicle specification	**NOTE:** After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the Diagnostic Trouble Code(s) ▪ Using the Jaguar Land Rover approved diagnostic equipment, check and update the car configuration file as necessary. Clear the Diagnostic Trouble Code(s) and retest
U2300-56	Central Configuration - Invalid/incomplete configuration	▪ Car configuration file mismatch with vehicle specification	**NOTE:** After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the Diagnostic Trouble Code(s) ▪ Using the Jaguar Land Rover approved diagnostic equipment, check and update the car configuration file as necessary. Clear the Diagnostic Trouble Code(s) and retest
U2300-64	Central Configuration - Signal plausibility failure	▪ Car configuration file mismatch with vehicle specification	**NOTE:** After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the Diagnostic Trouble Code(s) ▪ Using the Jaguar Land Rover approved diagnostic equipment, check and update the car configuration file as necessary. Clear the Diagnostic Trouble Code(s) and retest
U3000-44	Control Module - Data memory failure	▪ Image processing module internal failure	▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest. If the fault persists, install a new image processing module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U3000-96	Control Module - Component internal failure	**NOTE:** This DTC may be accompanied by the message 'Autonomous Emergency Braking (AEB)/Lane Departure Warning (LDW)/Forward alert not available' displayed on the instrument cluster ▪ Image Processing Module (IPMA) is not configured correctly ▪ Car configuration mismatch with the vehicle specification ▪ Image Processing Module (IPMA) internal failure	**NOTES:** ▪ This procedure requires the Jaguar Land Rover (JLR) approved diagnostic equipment. ▪ For more detailed description on how to diagnose this DTC refer to the workshop manual section, (413-09) Waring Devises or follow the link below ▪ For additional information, refer to: **Warning Devices** (413-09 Warning Devices, Diagnosis and Testing). ▪ Using the Jaguar Land Rover approved diagnostic equipment, reconfigure the Image Processing Module (IPMA) with the latest level software. Road test the vehicle and retest ▪ Using the Jaguar Land Rover approved diagnostic equipment, check and update the car configuration file as necessary ▪ If the fault persists, using the Jaguar Land Rover approved diagnostic equipment, reconfigure the Image Processing Module (IPMA) with the latest level software. Clear the Diagnostic Trouble Code(s), road test the vehicle and retest
U3000-97	Control Module - Component or system operation obstructed or blocked	▪ Image processing module camera obstructed	▪ Remove obstructions from in front of the image processing module
U3003-16	Battery Voltage - Circuit voltage below threshold	▪ Image processing module power or ground circuit open circuit, high resistance ▪ Battery/charging system fault	▪ Using the Jaguar Land Rover approved diagnostic equipment, check datalogger signal - ECU Power Supply Voltage (0xD112). Refer to the electrical circuit diagrams and check the image processing module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the relevant section of the workshop manual and test the battery and charging system
U3003-17	Battery Voltage - Circuit voltage above threshold	▪ Battery/charging system fault	▪ Using the Jaguar Land Rover approved diagnostic equipment, check datalogger signal - ECU Power Supply Voltage (0xD112). Refer to the relevant section of the workshop manual and test the battery and charging system

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